

18.06 Linear Algebra Lectures 23 & 24
Textbook Notes - Sections 6.3 & 6.4

• Section 6.3 Systems of Differential Equations

• Systems of DE

The system $\frac{du}{dt} = Au$ is solved by the same eigenvalue machinery as $u_{k+1} = Au_k$, but with e^{kt} replacing λ^k . Each eigenvector evolves independently as a pure exponential $e^{kt}x$, and the general solution is a combination of these.

• Solving $\frac{du}{dt} = Au$

The same three steps for difference equations:

1. Decompose write $u(0) = c_1x_1 + c_2x_2 + \dots + c_nx_n$
where x_i are eigenvectors
Equivalently $c = X^{-1}u(0)$

2. Evolve each eigenvector gets multiplied by its
growth factor e^{kt} (not λ_i^k)

3. Recombine

$$u(t) = c_1 e^{k_1 t} x_1 + c_2 e^{k_2 t} x_2 + \dots + c_n e^{k_n t} x_n$$

Why? if $Ax = \lambda x$, then $u(t) = e^{kt}x$ satisfies
 $\frac{du}{dt} = \lambda e^{kt}x = Ae^{kt}x = Au$

• Stability

The solution $u(t) \rightarrow 0$ as $t \rightarrow \infty$ if and only if every eigenvalue has negative real part.

For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, two conditions are sufficient:

$$\begin{cases} \rightarrow \text{trace} = a + d < 0 & (\text{sum of eigenvalues is negative}) \\ \rightarrow \det A = ad - bc > 0 & (\text{product of eigenvalues is positive}) \end{cases}$$

↳ Together these force both eigenvalues (real or complex) to have negative real parts.

• Behaviour by Eigenvalue type

- λ real and negative
- λ real and positive
- $\lambda = r \pm is$ complex

→ λ purely imaginary

- pure decay $e^{kt} \rightarrow 0$
- blowup
- $e^{rt}(\cos st + i \sin st)$
oscillation modulated by
growth ($r > 0$) or decay ($r < 0$)
- pure oscillation, no growth,
nor decay.

• Second-order Equations

A scalar equation $my'' + by' + ky = 0$ converts to a first-order system by setting $u = (y, y')^T$:

$$\frac{du}{dt} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} u$$

↳ Companion matrix $\rightarrow$ eigenvalues satisfy same characteristic equation $m\lambda^2 + b\lambda + k = 0$.

$\rightarrow$ Three damping regimes:

↳ $b^2 < 4mk \rightarrow$ complex λ , underdamping (oscillation with decay)

↳ $b^2 = 4mk \rightarrow$ repeated λ , critical damping

↳ $b^2 > 4mk \rightarrow$ real λ , overdamping (pure decay, no oscillation)

• The Matrix Exponential e^{At}

Defined by the infinite series:

$$e^{At} = I + At + \frac{1}{2}(At)^2 + \frac{1}{6}(At)^3 + \dots$$

Key Properties

• $\frac{d}{dt}(e^{At}) = Ae^{At}$, so $u(t) = e^{At}u(0)$ solves $\frac{du}{dt} = Au$ for any $u(0)$.

• If $A = X\Lambda X^{-1}$, then $e^{At} = X e^{\Lambda t} X^{-1}$ where $e^{\Lambda t} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t})$

• e^{At} is always invertible (inverse is e^{-At})

• If $Ax = \lambda x$, then $e^{At}x = e^{\lambda t}x$

Special Properties

1. e^{At} always invertible $\rightarrow$ eigenvalues $e^{\lambda t}$ are never zero.
2. When A is antisymmetric ($A^T = -A$), e^{At} is orthogonal (pure rotation).
3. When $(A - \lambda I)^2 = 0$ (repeated eigenvalue, deficient eigenvectors), the series produces $te^{\lambda t}$ terms.

◦ Symmetric Matrices - Section 6.4

Symmetric matrices have the strongest possible eigenvalue structure:

- ↳ all eigenvalues are real
- ↳ there is always a full set of orthonormal eigenvectors.
- ↳ Diagonalization $A = X\Lambda X^{-1}$ becomes $S = Q\Lambda Q^T$ (no inverse needed, just a transpose).

• The Spectral Theorem

Every real symmetric matrix $S = S^T$ satisfies:

$$S = Q\Lambda Q^T$$

where Λ contains real eigenvalues and Q is orthogonal ($Q^T = Q^{-1}$) with orthonormal eigenvectors as columns.

↳ (Guaranteed even with repeated eigenvalues)

→ Two key facts and their proofs

1. Eigenvalues are real → From $Sx = \lambda x$, take conjugates to get $S\bar{x} = \bar{\lambda}\bar{x}$

Then $\bar{x}^T Sx = \lambda \|x\|^2$ and also $\bar{x}^T Sx = \lambda \|x\|^2$

Since $\|x\|^2 \neq 0$, $\lambda = \bar{\lambda}$, forcing λ to be real.

2. Eigenvectors for distinct Eigenvalues are orthogonal

↳ From $Sx = \lambda_1 x$ and $Sy = \lambda_2 y$

$$\lambda_1 x^T y = (Sx)^T y = x^T S^T y = x^T Sy = \lambda_2 x^T y$$

↳ Since $\lambda_1 \neq \lambda_2$, this forces $x^T y = 0$

• Eigenvalues vs pivots

For symmetric matrices, the signs match:

$$\{ \text{number of positive eigenvalues} = \text{number of positive pivots} \}$$

↳ Connects two halves of applied Linear Algebra:

• $S = LDL^T$, where D contains the pivots

• As L moves continuously toward I , eigenvalues move continuously towards pivots (and cannot cross zero as S stays nonsingular), so signs are preserved

• The $S = Q\Lambda Q^T$ Decomposition as a Sum

The spectral decomposition can be written as a sum of rank-one matrices.

$$S = \lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T + \dots + \lambda_n q_n q_n^T$$

↳ Each $q_i q_i^T$ is a projection onto the eigenvector q_i , scaled by the eigenvalue λ_i .

• Special Matrix Types and their Eigenvalues

Type	Condition	Eigenvalue Property
Symmetric	$S^T = S$	All λ real
Antisymmetric	$A^T = -A$	All λ purely imaginary
Orthogonal	$Q^T = Q^{-1}$	All $ \lambda = 1$ (all eigenvalues lie on unit circle)

• Complex Eigenvalues of real matrices

For any real matrix (not necessarily symmetric), complex eigenvalues and eigenvectors always come in conjugate pairs:

$$\rightarrow \text{If } Ax = \lambda x, \text{ then } A\bar{x} = \bar{\lambda} \bar{x}$$

This is because taking the conjugate of $Ax = \lambda x$ and using the fact A is real gives the second equation.